

Chapter III

Image Enhancement In Frequency Domain

11 Aug 15N

12 marks

[6 hours]

Fourier Transform

Fourier transform is a mathematical tool to analyze and design linear system. It is used to reduce the number of calculation to a fraction.

It also helps to quantify the effects of digitizing system.

The application of Fourier transform is:

- Image Enhancement
- X-Ray
- Enhancement of TV transmission & video signal.

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1) 1-D Fourier Transform

Let $f(x)$ be a continuous real variable of x .

Then,

Fourier Transform = $F[f(x)] = F(u)$

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$

Inverse Fourier transform is

$$F^{-1}[F(u)] = f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} dx$$

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2) 2-D Fourier Transform

A 2D function $f(x, y)$ has 2D transform as:

$$F[f(x, y)] = F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux + vy)} dx dy$$

Inverse, Fourier transform is

$$F^{-1}[F(u, v)] = f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux + vy)} dx dy$$

Properties of Fourier transform

(i) Addition Theorem:

$$F[f(x, y) + g(x, y)] = F(u, v) + G(u, v)$$

(ii) Shift Theorem:

$$F[f(x-a, y-b)] = F(u, v) e^{-j2\pi(ua + vb)}$$

(iii) Similarity Theorem:

$$F[f(ax, by)] = \frac{1}{|ab|} F\left(\frac{u}{a}, \frac{v}{b}\right)$$

(iv) Convolution Theorem:

$$F[f(x,y) * g(x,y)] = F(u,v) * G(u,v)$$

(v) Rayleigh's Theorem:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x,y)|^2 dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |F(u,v)|^2 du dv$$

(vi) Evenness & Oddness:

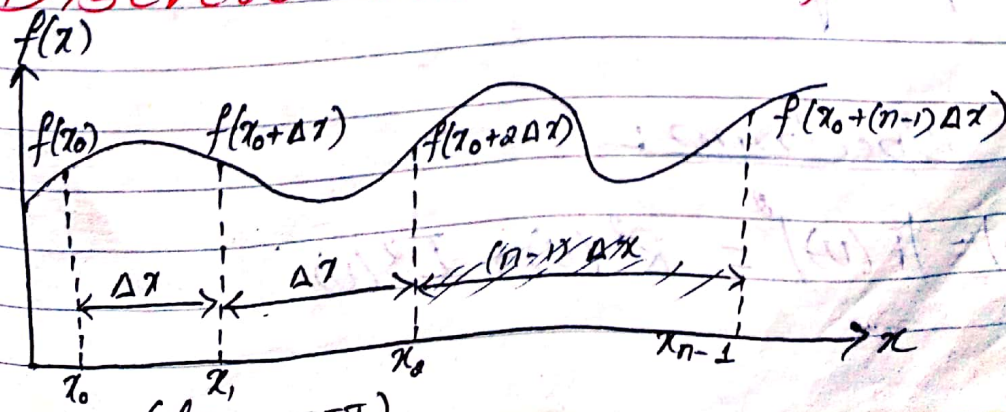
$$f_e(x) \Rightarrow f(x) = f(-x)$$

$$f_o(x) \Rightarrow f(-x) = -f(x)$$

(vii) Power Theorem:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) * g^*(x,y) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u,v) * G^*(u,v) du dv$$

Discrete Fourier Transform (DFT)



Here, $f(x)$ is a continuous function to be discretized by taking 'n' samples i.e. $\{f(x_0), f(x_0 + \Delta x), \dots, f(x_0 + (n-1)\Delta x)\}$.

It is defined as

$$F(u) = \sum_{x=0}^{n-1} f(x) e^{-j2\pi \frac{ux}{n}}$$

Inverse discrete Fourier transform is defined as:

$$f(x) = \sum_{u=0}^{n-1} F(u) e^{j2\pi \frac{ux}{n}}$$

These two given equations are discrete Fourier transform and it is used to calculate the following:

(i) Fourier Spectrum:

$$|F(u)| = \sqrt{R^2(u) + I^2(u)}$$

(ii) Power Spectrum:

$$|P(u)| = |F(u)|^2 = R^2(u) + I^2(u)$$

(iii) Phase angle spectrum : -

$$\phi = \tan^{-1} \left(\frac{I(u)}{R(u)} \right)$$

Where,

'u' is transform frequency variable
'x' is spatial or image variable

Discrete Fourier Transform (DFT) is most widely used orthogonal transform in the field of image processing & it can diagonalized circulant matrix.

However, in this transform, it takes n^2 multiplication and $n(n-1)$ addition to calculate 1D discrete fourier transform of 'n' data points.

Discrete Cosine Transform & Its Properties

Discrete Cosine Transform is a fourier related transform similar to DFT but it is only concerned with real number only.

It can be defined as:

$$DCT = C(u) = \alpha(u) \sum_{x=0}^{N-1} f(x) \cos\left(\frac{(2x+1)u\pi}{2N}\right)$$

where,

$$u = 0, 1, \dots, N-1$$

$$\alpha(u) = \sqrt{\frac{1}{N}} = \frac{1}{\sqrt{N}} \text{ for } u=0$$

$$= \frac{2}{\sqrt{N}} \text{ for } u=1, 2, \dots, N-1$$

Inverse DCT is given by:

$$IDCT = f(x)$$

$$= \sum_{u=0}^{N-1} \alpha(u) \cos\left(\frac{(2x+1)u\pi}{2N}\right)$$

$$\text{where, } u = 0, 1, 2, \dots, N-1$$

For 2D

$$DCT = C(u, v) = \alpha(u, v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos\left(\frac{(2x+1)u\pi}{2N}\right) \cos\left(\frac{(2y+1)v\pi}{2N}\right)$$

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where,

$$\alpha(u, v) = \frac{1}{N} \text{ for } u, v = 0$$

$$= \frac{\sqrt{2}}{N} \text{ for } u, v = 1, \dots, N-1$$

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$$\text{IDCT} = f(x, y)$$

$$= \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} \alpha(u, v) \cos\left(\frac{(2x+1)u\pi}{2N}\right) \cos\left(\frac{(2y+1)v\pi}{2N}\right)$$

where,

$$x = y = 0, 1, 2, \dots, N-1$$

Properties of DCT

- (i) The cosine transform is real and orthogonal.

$$C = C^* \Rightarrow C^{-1} = C^T$$

where,

C^* = Conjugate of C

C^T = Transpose of C

- (ii) The cosine transform is not real part of unitary DFT.

$$T^{-1} = T^{*T}$$

- (iii) It is a fast transform. The cosine transform of 'N' elements can be calculated in $O(N \log_2 N)$ operation through 'N' point FFT (Fast Fourier Transform).
- (iv) It has excellent energy compaction for highly correlated data.

Uses of DCT

DCT is used in image processing & signal processing especially for lossy data compression. Since, it has a strong energy compaction features.

It is also used for JPEG image compression, MPEG video compression and MJPEG video compression.

~~Imp~~ ~~Had~~ Hadamard Transform

Hadamard transform is a fast transform which is real, symmetric and orthogonal.

1-D hadamard transform can be implemented in $O(N \log_2 N)$ addition & subtraction.

It has only binary values i.e. 0, 1 or -1 in its kernel matrix. It is defined as:

$$H(u) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} f(x) (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$$

Kernel

for $N=2$,

$$N = 2^n \Rightarrow 2 = 2^n \Rightarrow n=1$$

where,

$$u = 0, 1, \dots, n-1$$

Now,

Hadamard matrix can be defined

as,

$$H_n = \frac{1}{\sqrt{N}} \begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -(H_{n-1}) \end{bmatrix}$$

where,

$$\frac{1}{\sqrt{N}} = \text{Sample value}$$

$$\begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -(H_{n-1}) \end{bmatrix} = \text{Indicate kernel}$$

$$H_0 = 1$$

Hadamard matrix at 1, $H_0 = 1$

$$H_n = \frac{1}{\sqrt{N}} \begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{bmatrix} \quad // \quad N=2 \quad \text{at } (0, 1)$$

$$H_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} H_0 & H_0 \\ H_0 & -H_0 \end{bmatrix}$$

$$H_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Also for $N=4$, $N=2^n \Rightarrow n=2$

$$H_2 = \frac{1}{\sqrt{4}} \begin{bmatrix} H_1 & H_1 \\ H_1 & -H_1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Properties of Hadamard transform

① It is real, symmetric and orthogonal.

$$\text{ie } H = H^* \Rightarrow H^{-1} = H^T$$

- (ii) It is a fast transform and 1-D Hadamard transform can be implemented in $O(N \log_2 N)$ addition & subtraction.
- (iii) It has only binary values i.e. 1 or -1 in its kernel matrix. No multiplication are required in the transform.
- (iv) It is used for digital image processing and digital signal processing.
- (v) It has good energy compaction for highly ~~correlated~~ correlated image.
- (vi) It is also used in digital hardware implementation of image processing algorithm.

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Haar Transform

Haar transform is the orthogonal transform which is derived from the Haar matrix.

It is also a fast transform and can be implemented in $O(N \log_2 N)$ operation.

The main advantage of this transform is that

It samples the input data sequence into fine resolution and takes the difference between the adjacent pairs.

The resolution increases by a power of 2 and as a result, in the transform domain differential energy is more & more localized.

It only exist for $n \cdot N = 2^n$ where, n is an integer.

Haar transform is defined as:

$$H_T = H_k(x) \quad \text{where, } k = 0, 1, \dots, N-1$$

k is given as:

$$k = 2^p + q - 1$$

and

$$0 \leq p \leq (n-1)$$

$$1 \leq q \leq 2^p \quad \text{for } p \neq 0$$

$$q = 0, 1 \quad \text{for } p = 0$$

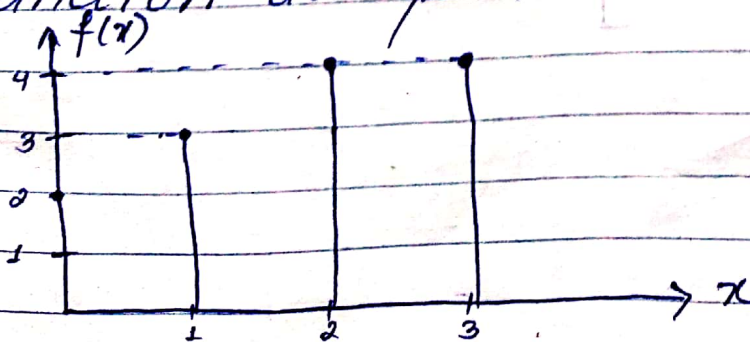
✓ Properties of Haar Transform

- (i) It is symmetric, separable, unitary transform that uses Haar function for its basis.
- (ii) It is orthogonal and real i.e.,

$$T^{-1} = T^T \Rightarrow H_r = H_r^*$$
- (iii) It is a fast transform and can be implemented in $O(N)$ operation where N is the number of samples.
- (iv) It exists for $N=2^n$ where n is an integer.
- (v) It has poor energy compaction property.

Numericals

Q Find the Fourier spectrum of the function at point 0 and 1 for given figure



$$\begin{aligned}
 f(0) &= 2 \\
 f(1) &= 3 \\
 f(2) &= 4 \\
 f(3) &= 4
 \end{aligned}$$

Fast Fourier Transform (FFT)

The Fast Fourier Transform is a discrete Fourier transform algorithm which reduces the number of computations needed for N points from $2N^2$ to $2N \log_2 N$.

An FFT rapidly computes the transformations by factorizing DFT matrix into a product of sparse (mostly zero) factors.

As a result, it manages to reduce the complexity of computing the DFT from $O(n^2)$ to $O(n \log_2 n)$, where n is the data size.

The FFT algorithm generally fall into two classes:

- decimation in time
- decimation in frequency.

Coley-Tukey FFT algorithm

The basic idea for FFT (decimation in time) is to break up a transform of length N in two transforms of length $N/2$ using the identity as:

$$\sum_{n=0}^{N-1} a_n e^{-\frac{2\pi i n k}{N}} = \sum_{n=0}^{N/2-1} a_{2n} e^{-\frac{2\pi i (2n) k}{N}} + \sum_{n=0}^{N/2-1} a_{2n+1} e^{-\frac{2\pi i (2n+1) k}{N}}$$

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$$= \sum_{n=0}^{N/2-1} a_n^{\text{even}} e^{-\frac{2\pi i k}{N/2}} + \left(\sum_{n=0}^{N/2-1} a_n^{\text{odd}} e^{-\frac{2\pi i k}{N/2}} \right) e^{-\frac{2\pi i k}{N}}$$

i.e., $X_k = E_k + e^{-\frac{2\pi i k}{N}} O_k$

It is sometime called Danielson Lanczos Lemma.

The easiest way to visualize this procedure is perhaps via the Fourier matrix.

Applications

- (i) Fast large integer & polynomial multiplication
- (ii) Filtering algorithms
- (iii) Solving difference equations
- (iv) Efficient matrix-vector multiplication